YANQING LIU

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RESEARCH FOCUS

My research interests lie at the intersection of human-AI interaction, multimodal sensing and adaptive systems. I study how interaction, conversational and physiological data can reveal users' momentary cognitive and emotional states, and how AI systems can adapt to them to better support people's learning, work, decision-making and well-being.

EDUCATION

University of New South Wales **Sydney, Australia | 2025 – 2026**

Graduate Certificate in Information Technology (exit award from the Master of IT) | GPA: 90.333/100, High Distinction

Coursework: *Interactive Design, Data Structure and Algorithm, Web Front-End Programming*

Columbia University **New York, USA | 2021 – 2023**

M.S. in Applied Analytics | GPA: 4.132/4.3

Coursework: *Research Design, Applied Text and Natural Language Analytics, Storytelling with Data*

Macau University of Science and Technology **Macau, China | 2016 – 2020**

Bachelor of Applied Economics | Major GPA: 3.86/4.0 | Rank: 2/882, School of Business

Coursework: *Advanced Mathematics, Linear Algebra, Statistics, Behavioral Economics, Game Theory*

PUBLICATIONS & PRESENTATIONS

Liu, Y. "From Sensing to Adaptation: A Scoping Review of Physiological Indicators of Learner States for Adaptive Learning Support." *HCI International 2027 (HCII 2027)*, Springer. [Accepted; full paper in preparation]

Liu, Y. "Mindful-Scroll: Linking Interaction Logs to Self-Reported Mental Health and Well-Being, and In-App Interventions." *Digital Mental Health International Conference 2026*, Hong Kong, China, June 2026. [Presentation]

Liu, Y., Zhou, W., Zhu, Y., Zhao, X., & Jin, Y. "Negotiating Reflective Demand: LLM-Mediated Reflection on Everyday Short-Form Video Use." *CHI 2027*. [Under Review]

Liu, Y., Zhu, Y., & Jin, Y. "Exploring Candidate Signals of Perceived Stress and Anxiety in Short-Form Video Use: Scrolling Logs and Use-Specific Appraisals." In *DMH Conference Book: Maximizing the Real-World Potential of Digital Mental Health*. Springer Nature. [Under Review]

Liu, Y., Zhu, Y., & Jin, Y. "Beyond Social-Media Posts: Conversational Reflection Signals of Daily Perceived Stress and Anxiety in Short-Form Video Use." [Manuscript in revision for journal resubmission]

Wang, Y., **Liu, Y.**, & Ray, L. C. "'Like Fire Alarm Promotes Waking Up': How Humor Can Affect User Engagement with Misinformation on Social Media." *CHI 2027 Late-Breaking Work*. [In Preparation]

Deng, W., **Liu, Y.**, Chen, Y., Zhang, H., Li, Y., Mi, H., & Yu, C. "LLM-Based Simulation to Support Student Academic and Life Supporters (Fudaoyuan) at Chinese Universities." *CHI 2027 Late-Breaking Work*. [In Preparation]

RESEARCH EXPERIENCE

Research Collaborator **Singapore Management University** **09/2026 – Present**

Advisor: Prof. Fiona Fui-Hoon Nah | School of Computing and Information Systems

Lead a scoping review of physiological learner-state indicators for adaptive learning (accepted at HCII 2027).

- Designed the protocol (JBI, PRISMA-ScR) and an iteratively tested Scopus search (5,211 unique records).
- Conducted a 200-record screening pilot, finding 44 of 51 retained studies stopped at association or detection without adaptation, and 15 inferred states from prior literature without validation in their own data.

Research Fellow (Full-Time) **Duke Kunshan University** **02/2026 – Present**

Advisor: Prof. Yucheng Jin | Hybrid Interactive Intelligence (HII) Lab

Lead Mindful Scroll, a mixed-methods project on short-form video use, well-being and LLM-mediated reflection.

- Ran a pilot survey (n = 32), identifying a four-trace behavioral usage cluster linked to stress and anxiety.
- Independently built Mindful Scroll, a React-based, Dockerized short-form video platform integrating behavioral logging, well-being self-reports, and LLM-mediated reflection dialogues.
- Ran a 14-day field study with 41 participants, collecting 44,420 video views and 113,233 log events.
- Developed an analysis pipeline combining GPT-assisted seeded topic modeling, predictive modeling, and longitudinal analyses linking micro-interactions and conversation patterns to stress/anxiety signals.
- Conducted 12 participant interviews to examine the perceived benefits and burdens of reflection, informing the design of interactions that let users negotiate reflection timing, depth, and interpretation.

Summer Research Intern **City University of Hong Kong** **06/2026 – 08/2026**

Advisors: Prof. Ray LC and Prof. Fiona Fui-Hoon Nah (co-advisor) | Studio for Narrative Spaces

Developed an experimental interface to study how comment conditions shape responses to misinformation.

- Led front-end development of a simulated social media feed for a between-subjects experiment, with randomized conditions and logging of share, repost, report and skip decisions.
- Led the quantitative analysis of 1,200 post-level decisions from 60 participants using GEE models.
- Found that humorous comments did not reliably change actions, whereas believing a post false raised reporting odds more than eightfold, and co-authored a CHI 2027 Late-Breaking Work manuscript.

Research Collaborator **Tsinghua University** **09/2025 – Present**

Advisor: Dr. Jie Cai | Pervasive HCI Group

Analyzed conversations in CoSim, an LLM simulation for training university student academic and life supporters.

- Built a trajectory-then-thematic pipeline across 10 supporters and 19 sessions to trace emotion, self-disclosure, and resistance across turns and contrast facilitative versus hindering sessions.
- Identified communication patterns, surfacing supporters' blind spots in emotionally sensitive conversations and informing simulation-based feedback design.
- Contributed to the mixed-methods analysis and co-authored a CHI 2027 Late-Breaking Work manuscript.

Course Research Projects **Columbia University** **2022**

Instructors: Prof. Kitty Kay Chan and Prof. Cesar Koirala | Industry Partner: Sandler Training Company

- Analyzed 200,000+ Yelp reviews with text mining, sentiment analysis and decision trees to identify pandemic-related themes and their links to ratings.
- Analyzed 479 client surveys with clustering, classification and random forest models, segmenting clients by revenue, growth and sentiment to identify high-value groups.

WORK EXPERIENCE

Frontend Developer Intern **Beijing Puppy Agent Technology Co., Ltd.** **08/2024 – 12/2024**

- Developed an enterprise RAG-based workflow platform, designing a hybrid ReactFlow state management architecture, cutting workflow creation time from 2 hours to 30 minutes.
- Built a node framework with automated edge generation and integrated Monaco Editor for rich-text editing.

SKILLS

Technical: Python • R • SQL • JavaScript • React • Docker • LLM APIs • Figma • MATLAB • Stata

Methods: Mixed methods • Thematic analysis • Topic modeling • Predictive modeling • A/B experiments

Languages: Mandarin (native) • English (fluent) | TOEFL iBT 104 • GMAT 690

AWARDS & HONORS

Australia's Global University Award, UNSW	2025
Dean's Scholarship, MUST	2019
Luso International Banking Ltd. Scholarship, MUST	2019
Nam Kwong (Group) Company Limited Scholarship, MUST	2018